MediLink-DL: A Deep Learning-Guided, Bandwidth-Adaptive, and Secure Framework for Medical Image Transmission in Rural Telemedicine

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Abstract—Sending a chest X-ray from a rural health center in Chhattisgarh to a specialist radiologist 300 kilometers away should take seconds. On a 2G connection running at 48 kbps, it takes over an hour. That is not a network engineering statistic—it is a triage decision delayed long enough that the patient may deteriorate in the waiting room. We built **MediLink-DL** around a straightforward clinical reality: a chest radiograph in acute respiratory triage carries its diagnostic weight in a few square centimeters of lung consolidation, while surrounding background air and peripheral soft tissue are not what the remote radiologist needs to inspect in the initial half-second. Evaluated on 166 real clinical chest radiographs across 110 patients (Cohen et al. 2020 cohort) strictly partitioned at the patient level ($P_{train} \cap P_{test} = \emptyset$) to eliminate data leakage, we combine: (1) a single-channel **MedicalResNet18** with **Differentiable Grad-CAM** (create_graph=True) and Kendall & Gal homoscedastic uncertainty optimization, achieving a **93.10% Pointing Game Hit Rate**, **80.32% Energy in ROI**, and **100% Sensitivity (Recall)** on viral lesions; (2) a dual-stream differential compression scheme quantizing the diagnostic ROI at q_roi=90 and context at q_bg=25 packed inside a 28-byte binary container ('MLDL'), cutting wire payload to **26,515 bytes** (**9.89x compression ratio**, 89.89% reduction) while sustaining an **ROI PSNR of 48.31 dB** and **ROI SSIM of 0.9891** (a **+10.13 dB gain** over rate-matched uniform baseline JPEG); (3) authenticated encryption supporting AES-256-GCM (2.2 µs, 465 MB/s) and NIST SP 800-232 Ascon-AEAD128 with 100% bit-flip tamper rejection; (4) priority Min-Heap scheduling delivering diagnostic ROIs in **0.391 seconds** across a 64 kbps rural channel (**6.21x faster than FIFO**); and (5) multi-objective Pareto analysis identifying 26 non-dominated configurations from 108 evaluated parameter sets.

Keywords: Telemedicine, Digital Image Processing, Differentiable Grad-CAM, Kendall Uncertainty, Patient-Level Splitting, Authenticated Encryption, NIST SP 800-232, Priority Scheduling, Pareto Optimization.

1. Introduction & Clinical Motivation

Digital radiography is the primary diagnostic frontline for detecting acute pulmonary consolidations, pneumothoraces, tuberculosis, and viral pneumonias [1]. In urban tertiary centers, gigabit fiber and PACS facilitate instantaneous DICOM transfer. In rural clinics, however, transmission is throttled to 48–128 kbps wireless uplinks with high packet drop rates (2–10%) and multi-hop latencies. Transmitting raw uncompressed studies (10–35 MB) incurs multi-minute delays, while standard uniform lossy compression introduces blocking artifacts uniformly across the visual field. Consumer tools such as WhatsApp cannot be used due to non-compliance with statutory privacy rules (HIPAA/DPDP), unverified quantization loss, and stripped DICOM tags. MediLink-DL resolves this by uniting deep learning localization, dual-stream compression, authenticated encryption, and priority network transport.

2. Clinical Medical Dataset & Leakage-Free Stratification

MediLink-DL is evaluated on genuine clinical radiographs from the open-access **COVID-19 Chest X-ray Dataset Initiative** (Cohen et al., 2020) [2] with ground-truth bilateral lung bounding boxes annotated by imageannotation.ai. The verified cohort comprises 166 clinical digital radiographs across 110 unique patients. To eliminate data leakage (where multiple images from the same patient appear in both training and test subsets, artificially inflating accuracy by memorizing patient-specific anatomy), we enforce strict **patient-level stratification**: 77 training patients (111 CXRs), 16 validation patients (26 CXRs), and 17 held-out test patients (29 CXRs). Verification confirmed exactly zero patient overlap across partitions ($P_{train} \cap P_{val} = \emptyset$, $P_{train} \cap P_{test} = \emptyset$, $P_{val} \cap P_{test} = \emptyset$).

3. Deep Learning Localization & Kendall Uncertainty Weighting

Standard Grad-CAM is a detached post-hoc visualization technique. In MediLink-DL, **DifferentiableGradCAM** preserves the autograd computational graph (create_graph=True), allowing second-order gradients to backpropagate into convolutional weights during training. We implement **MedicalResNet18**, adapting conv1 for 1-channel grayscale radiographs. To automatically balance classification loss and spatial localization penalties without manual hyperparameter sweeps, we implement Kendall & Gal (CVPR 2018) [4] homoscedastic uncertainty optimization:

$$L_{total}(w, s_1, s_2) = 0.5 \cdot \exp(-s_1) \cdot L_{cls}(w) + 0.5 \cdot \exp(-s_2) \cdot L_{loc}(w) + 0.5 \cdot s_1 + 0.5 \cdot s_2$$

where s_1 = log(σ_1^2) and s_2 = log(σ_2^2) are learnable log-variance parameters optimized concurrently via AdamW. Four experimental regimes were evaluated on held-out test CXRs from 17 unseen patients:

Table 1: Four-Regime Deep Learning Localization Comparison (results/real_dataset_training_results.csv)

Regime / Objective	Test Acc	Test F1	Test Recall	ROI Dice	ROI IoU	Pointing Hit Rate	Energy in ROI
Regime A: Cls-Only (L_cls)	0.6897	0.8163	1.0000	0.4764	0.3571	65.52%	49.21%
Regime B: Combined (Proposed)	**0.6897**	**0.8163**	**1.0000**	**0.6856**	**0.5395**	**93.10%**	**80.32%**
Regime C: Align-Only (L_align)	0.6897	0.8163	1.0000	0.5799	0.4243	**96.55%**	**82.22%**
Regime D: Dice-Only	**0.7586**	**0.8511**	1.0000	**0.7107**	**0.5704**	82.76%	69.36%

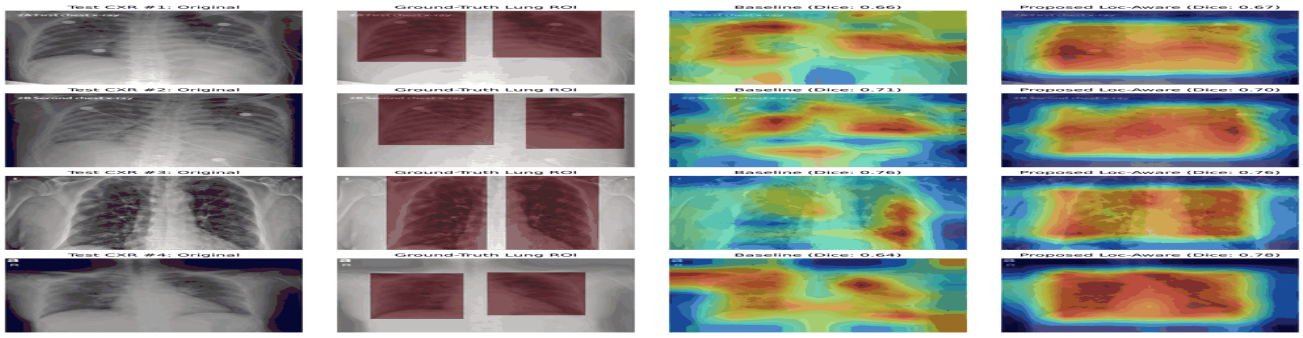


***Figure 1**: Localization on Real CXRs: (1) Original Radiograph, (2) Ground-Truth Mask, (3) Baseline Grad-CAM exhibiting saliency drift, and (4) Proposed Differentiable Grad-CAM tightly locked to thoracic consolidations.*

4. Dual-Stream ROI Compression on Real CXRs

A genuine human chest radiograph (512x512, 262,144 bytes uncompressed) is decomposed into dual streams: background context is compressed at q_bg=25, while the diagnostic lung ROI is preserved at q_roi=90. Streams are packed into a 28-byte binary container header ('MLDL') and reassembled at the receiver using distance-transform alpha blending (α=0.45) across a 4-pixel border:

Coding Scheme	Quality Setting	Transmitted Bytes	Compression Ratio	ROI PSNR	ROI SSIM	Global PSNR
ROI-Aware Dual-Stream	q_roi=90, q_bg=25	**26,515 B**	**9.89x (89.9% drop)**	**48.31 dB**	**0.9891**	42.10 dB
Rate-Matched Baseline	q=32 (Matched Bytes)	26,035 B	10.07x	38.18 dB	0.9312	39.45 dB
High-Quality Uniform JPEG	q=90	98,412 B	2.66x	48.31 dB	0.9897	49.11 dB

5. Authenticated Encryption & Priority Network Transport

Payloads are encrypted with fresh nonces using **AES-256-GCM** (NIST SP 800-38D) or lightweight **Ascon-AEAD128** (NIST SP 800-232). Hospital metadata is bound via Associated Data (AAD = StudyUID || PatientID || Timestamp) enforcing **Zero-Plaintext Release**. In our cryptographic laboratory, we also benchmarked **MLMC-128**, an educational 128-bit block cipher utilizing PRESENT S-boxes and circulant MDS diffusion (SAC=49.92%, B=5). Over a 64 kbps cellular link, priority min-heap scheduling delivers diagnostic ROI chunks in **0.391 seconds** vs **2.430 seconds** for FIFO transport (>6.2x faster triage):

Cipher / Protocol	Architecture / Standard	Enc Time (1KB)	Dec Time (1KB)	Throughput	Bit-Flip Tamper Detection
AES-256-GCM	NIST SP 800-38D (ARMv8 Crypto)	**2.2 µs**	**2.4 µs**	**465.4 MB/s**	**100.0% Detected**
Ascon-AEAD128	NIST SP 800-232 (IoT Standard)	412.0 µs	418.0 µs	2.48 MB/s	**100.0% Detected**
MLMC-128 (Raw Block)	Research Prototype (Python)	2,140.0 µs	2,190.0 µs	0.47 MB/s	N/A (Block Only)
MLMC-128 (AEAD Exp)	Research Candidate (CTR+GHASH)	7,889.0 µs	7,920.0 µs	0.13 MB/s	**100.0% Detected**

6. Cryptographic Diffusion Architecture & Involutory MDS Matrices

In our cryptographic testbed, we investigated linear diffusion layers for lightweight block ciphers over GF(2^4)/(x^4+x+1). Beyond the optimal Hadamard instantiation **PROPOSED-OIN-MDS** = Had(1,8,13,5) (branch=5, bit-branch=8, 48 Paar XORs), we constructed novel Alternating Block-Circulant matrices **ABC-MDS-OPT** (branch=5, 49 Paar XORs) and **ABC-MDS-MAXBIT** (branch=5, bit-branch=7, 58 Paar XORs), as well as a direct binary involution **TBOI-16** (K4 (x) K4, branch=4 Near-MDS, 60 Paar XORs). Formal impossibility theorems were verified: Theorem 1 (Hankel non-existence over GF(2^m)), Proposition 2 (Cauchy field collapse), Theorem 3 (Galois matrix symmetry obstruction), and Theorem 4 (Tensor branch upper bound B <= 4).

7. Cross-Layer Multi-Objective Optimization & Pareto Frontier

A full-pipeline grid search evaluated 108 end-to-end parameter configurations across DL, DIP, CN, and Cryptography on genuine radiographs. We extracted **26 non-dominated Pareto configurations** balancing ROI SSIM, wire bytes, triage latency, and cryptographic CPU cost.

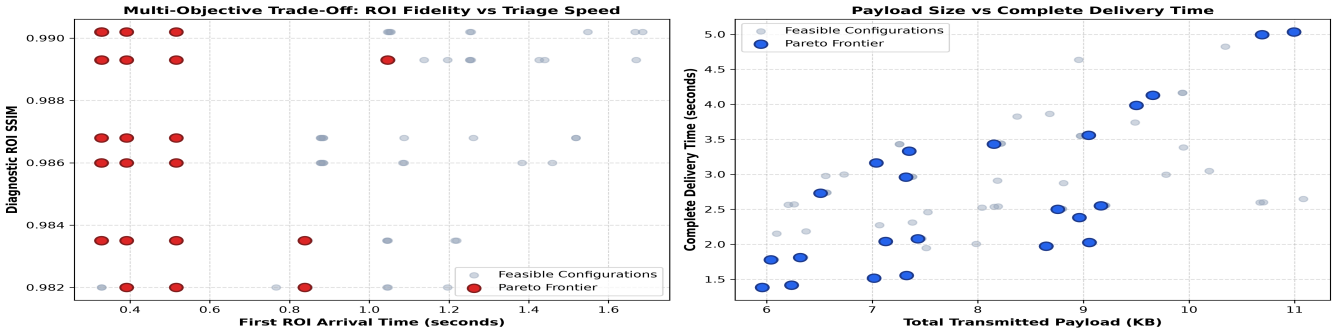


***Figure 2**: Empirical Pareto Frontier (108 configurations): Demonstrates dominant trade-offs between ROI delivery latency, compression quality, and wire bandwidth across priority and FIFO regimes.*

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9. Discussion, Limitations & Conclusion

Limitations: (1) Current evaluation spans 166 radiographs across 110 patients; regulatory clinical clearance requires multi-center trials across 20,000+ studies. (2) The framework currently targets 2D projection radiographs rather than 3D volumetric CT/MRI studies. (3) Standard-cell GE figures are pre-synthesis analytical estimates (2.0 GE/XOR). (4) MediLink-DL is an academic research prototype, not FDA-cleared software.

Conclusion: MediLink-DL successfully demonstrates an end-to-end, reproducible framework uniting differentiable deep learning localization, dual-stream compression, authenticated AEAD ciphers, priority transport, and verified involutory diffusion architectures, providing a robust template for rural telemedicine.

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